

COMPUTED BOUNDING COCHAINS IN THE PILLOWCASE: THE PRETZEL KNOTS $P(-2, 3, 5)$ AND $P(-2, 3, 7)$

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ABSTRACT. On the pillowcase (symplectic) side of the knot Atiyah–Floer conjecture, the naive Lagrangian–Floer homology of a tangle decomposition is not a knot invariant; the conjectural repair of Cazassus–Herald–Kirk–Kotelskiy is a *bounding cochain* b supported on the self-intersections of the immersed curves and satisfying a mod-2 Maurer–Cartan equation. Smith proved that for the pretzel knot $P(-2, 3, 5) = T(3, 5)$ one such bounding cochain must be nonzero, but produced no value, and to date no nonzero pillowcase bounding cochain has been computed. Working within the immersed-curve combinatorial model of Herald–Kirk and Smith, we reconstruct the relevant Lagrangians of the pretzel family $P(-2, 3, 2k + 1)$ from the quaternionic representation theory—reproducing, for $P(-2, 3, 5)$, Smith’s count of nine generators, two immersed bigons, and rank-5 Floer homology as a validation gate—and then compute the bounding cochains of its two smallest members explicitly. For $P(-2, 3, 5)$,

$$b = s_A + s_B,$$

the sum of two self-intersections of $\mathcal{R}_t(Q_{1/3} + Q_{1/5})$, one on each pillowcase seam; it raises the rank from 5 to $7 = \text{rank } I^\natural$ by a single immersed *quadrilateral* (μ^3) that cancels one bigon. For $P(-2, 3, 7)$, b is a *single* self-intersection, raising the rank from 7 to $9 = \text{rank } I^\natural$ by an immersed *triangle* (μ^2). In each case b is the unique minimal bounding cochain matching $I^\natural$, and cancels exactly one bigon (+2 to the homology rank); the two examples realize the same correction through different polygon orders. Both values are stable under change of perturbation. These are, to our knowledge, the first explicitly computed nonzero bounding cochains in pillowcase Floer theory. We separate carefully what is proved, what is computed within the model, and what remains conjectural (the faithfulness of the model), and we close with the computed data and open questions for the rest of the family, where the pattern does not obviously persist.

1. INTRODUCTION

1.1. The pillowcase, and why a bounding cochain is needed. Let $K \subset S^3$ be a knot presented as a union of two tangles, $K = (B^3, T_1) \cup_{(S^2, 4)} (B^3, T_2)$, glued along a four-punctured sphere. The traceless $\text{SU}(2)$ character variety of $(S^2, 4)$ is the *pillowcase*

$$P = (\mathbb{R}/2\pi\mathbb{Z})^2 / \iota, \quad \iota(\gamma, \theta) = (-\gamma, -\theta),$$

a two-sphere with four $\mathbb{Z}/2$ orbifold points (the *corners*) at $(\gamma, \theta) \in \{0, \pi\}^2$ and symplectic form descending from $d\gamma \wedge d\theta$ on the smooth locus P^* [9, 4]. Each tangle T_i determines an immersed Lagrangian $\mathcal{R}_{\pi_i}(T_i) \looparrowright P^*$ (its perturbed traceless character variety), and the intersection points $\mathcal{R}_{\pi_1}(T_1) \cap \mathcal{R}_{\pi_2}(T_2)$ are the traceless representations of $\pi_1(S^3 \setminus K)$ —the generators of the reduced singular instanton chain complex of Kronheimer–Mrowka [7, 8]. Hedden, Herald and Kirk [4, 5] proposed building a knot invariant as the Lagrangian–Floer homology $HF(\mathcal{R}_{\pi_1}(T_1), \mathcal{R}_{\pi_2}(T_2))$ of this pair, an Atiyah–Floer counterpart of $I^\natural(K)$.

This naive homology is *not* a knot invariant: it changes under the tangle moves that fix K , because of figure-eight bubbling at the immersed curves’ self-intersections. Cazassus, Herald, Kirk and Kotelskiy [2] identified the repair. To each tangle one attaches a *bounding cochain*

$b \in CF(\mathcal{R}(T), \mathcal{R}(T))$ —a formal $\mathbb{F}_2$ -combination of the self-intersection points—subject to the Maurer–Cartan equation

$$\sum_{k \geq 0} \mu^k(b, \dots, b) = 0, \quad (1)$$

where μ^k counts immersed $(k+1)$ -gons and μ^1 is the Floer differential. The bounding cochain deforms the differential of $CF(\mathcal{R}(T_1), \mathcal{R}(T_2))$ into ∂_b , whose homology $HF((\mathcal{R}(T_1), b_1), (\mathcal{R}(T_2), b_2))$ is conjecturally the knot invariant and equal to $I^\natural(K)$ [2, Conj. 1.1].

1.2. The problem and the results. The bounding cochain is the analytic heart of the theory, and the hardest part to make explicit. It is inert precisely where the curves are embedded: for two-bridge knots the Lagrangians are straight arcs with no self-intersections, $b = 0$, and the naive homology is already $I^\natural$ [4, 5]. Off that family it is genuinely nonzero. Two facts are known. For the torus knot $T(4, 5)$, Cazassus–Herald–Kirk–Kotelskiy exhibit a bounding cochain on the *earring* tangle that corrects the rank from 9 to the true 7 [2, §10]. For the pretzel knot $P(-2, 3, 5)$, Smith [10, Thm. 5.9] proves that the bounding cochain b of the summand tangle $T_2 = Q_{1/3} + Q_{1/5}$ must be nonzero—by computing that the naive homology has rank 5 while $I^\natural = 7$ —but he stops there: no value, and no algorithm, is given. In both cases the underlying combinatorial data was, until now, the extent of what had been extracted; *no nonzero pillowcase bounding cochain has been computed*.

We compute two of them. Both are members of the pretzel family

$$P(-2, 3, 2k+1) = \text{num}(Q_{-1/2} + Q_{1/3} + Q_{1/(2k+1)}), \quad \gcd(2k+1, 3) = 1, \quad (2)$$

with the decomposition $T_1 = \widehat{Q_{-1/2}}$ (to which the earring is added, so its bounding cochain vanishes) and $T_2 = Q_{1/3} + Q_{1/(2k+1)}$; the summand $Q_{1/3} + Q_{1/(2k+1)}$ carries the interesting cochain. Its smallest members are $P(-2, 3, 5) = T(3, 5)$ ($k = 2$) and $P(-2, 3, 7)$ ($k = 3$). Throughout, “computed within the model” means computed in the immersed-curve combinatorial Floer theory of [6, 10], in which Lagrangian–Floer polygon counts are winding-number computations for piecewise-linear immersed curves in P ; the faithfulness of that model is the standing conjecture of the subject (§5).

Theorem 1.1. *Write $\mathcal{R}^\natural(T_1)$ (red) for the earring figure-eight of $\widehat{Q_{-1/2}}$ and $\mathcal{R}_t(T_2)$ (blue) for the perturbed Conway sum $\mathcal{R}_t(Q_{1/3} + Q_{1/(2k+1)})$. Within the combinatorial model:*

- (i) ($P(-2, 3, 5)$, $k = 2$.) *Red and blue have nine intersection points and exactly two immersed bigons with distinct vertices, so the naive rank is $\text{rank } HF((\mathcal{R}^\natural, 0), (\mathcal{R}_t, 0)) = 5$ (Smith’s gate, reproduced). The unique bounding cochain of minimal support with deformed rank $I^\natural = 7$ is*

$$\boxed{b = s_A + s_B}, \quad s_A \approx (\gamma, \theta) = (0.03, 1.27), \quad s_B \approx (3.06, 4.98),$$

two self-intersections of blue, one on each seam $\gamma \in \{0, \pi\}$; it raises the rank $5 \rightarrow 7$ by a single immersed quadrilateral (μ^3) that cancels one bigon.

- (ii) ($P(-2, 3, 7)$, $k = 3$.) *Red and blue have thirteen intersection points and three bigons, naive rank $\text{rank } HF((\mathcal{R}^\natural, 0), (\mathcal{R}_t, 0)) = 7$. The unique bounding cochain of minimal support with deformed rank $I^\natural = 9$ is a single self-intersection $s \approx (0.05, 5.41)$; it raises the rank $7 \rightarrow 9$ by an immersed triangle (μ^2) that cancels one bigon.*

In both cases the correction cancels exactly one bigon (+2 to the homology rank) and matches $I^\natural$; the two examples realize this through different polygon orders. Each value is stable under change of the perturbation parameters.

The blue Lagrangians are unobstructed— $\mu^0 = 0$, no teardrops—so (1) is a genuine equation for b and not merely a formal series; we record this (Proposition 3.1) as the standard degree phenomenon for curves in surfaces. For $P(-2, 3, 5)$ the low self-Floer products μ^1, μ^2 also vanish, so Maurer–Cartan is vacuous on small support and b is pinned by the rank condition alone; for $P(-2, 3, 7)$ the self-differential μ^1 is nonzero, but the support crossing of b is μ^1 -closed, so (1) still holds. In each case the support crossings carry no other polygons that would spoil the surgery, which is why the answer is so small (two crossings, resp. one).

1.3. Two mechanisms for one correction. It is worth isolating what the two examples share and where they differ. Both are *Conway-sum* bounding cochains, and both *cancel* a bigon: the naive rank (5, resp. 7) is *below* the target (7, resp. 9), so b must *destroy* a differential—remove an existing bigon—to raise the homology rank by 2. This is the opposite of the only prior example: for $T(4, 5)$ the bounding cochain lives on the *earring* and *creates* a differential, lowering the rank $9 \rightarrow 7$ [2]. Cancelling is the more rigid direction: a canceling polygon must land on the precise chain-level entry of an existing bigon. Where the two examples differ is the *order* of that polygon. For $P(-2, 3, 5)$ no triangle reaches either bigon pair, so the cancellation is a quadrilateral through two crossings; for $P(-2, 3, 7)$ a triangle does reach a bigon, so a single crossing suffices. The +2 correction is uniform; the polygon that carries it is not.

1.4. Method and honesty. We did not take the immersed curves from [10] as input. Both Lagrangians were reconstructed from first principles—the quaternionic representation theory of the tangles, Smith’s fiber-product description of the Conway sum, his cut-and-paste resolution of the seam circles, and the Herald–Kirk earring—and only then validated against the numerical output of [10]: the convention derivations, the seam fiber-circle arc, the resolution of the worked example $\mathcal{R}_t(Q_{1/2} + Q_{-1/3})$, and the nine generators, two bigons and rank 5 of Theorem 1.1(i) all reproduce Smith’s published figures and numbers. Only after that gate did we compute the bounding cochains, which Smith does not.

We are explicit about the three registers, as in the companion paper [11]. The construction of the Lagrangians, the deformed-differential formalism, and the unobstructedness of Proposition 3.1 are (i) reproduced from [4, 5, 2, 6, 10, 1], independently re-derived and re-implemented here. The polygon and Maurer–Cartan counts, the rank computations, and the identification of the bounding cochains are (ii) computer computations, carried out by a short self-contained program included with the source and validated against [10] at every prior stage; we report them as such. The values are the values of the nonzero bounding cochains (iii) *under* the standing conjecture that the combinatorial model computes the Floer-theoretic invariant—the same hypothesis under which every computation in this subject, including [2, 10], is stated. We prove no new theorem about the analytic Atiyah–Floer correspondence, and we make no claim about the family (2) beyond its two smallest members (§5.3).

§2 recalls the pillowcase and assembles the family’s Lagrangians, with the two validation gates. §3 states the combinatorial Floer package—winding-number counting, the lift bookkeeping, the deformed differential, and unobstructedness. §4 is the computation: the Maurer–Cartan data, the quadrilateral of $P(-2, 3, 5)$, and the triangle of $P(-2, 3, 7)$. §5 discusses robustness, the standing conjecture, and the family beyond $k = 3$.

Acknowledgements. The immersed-curve model and the geometry we reconstruct are due to Hedden–Herald–Kirk [4, 5], Cazassus–Herald–Kirk–Kotelskiy [2], Herald–Kirk [6], and Smith [10]; we have tried to attribute each ingredient precisely.

2. THE LAGRANGIANS OF THE FAMILY $P(-2, 3, 2k + 1)$

2.1. The pillowcase and immersed curves. We use $SU(2)$ as the unit quaternions; a traceless element is a purely imaginary unit, i.e. a point of the 2-sphere $S \subset \text{Im } \mathbb{H}$. The traceless character variety of the four-punctured sphere is the pillowcase $P = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota$ of the introduction, coordinatized by (γ, θ) with $\iota(\gamma, \theta) = (-\gamma, -\theta)$ and corners $P^c = \{0, \pi\}^2$ [9, 4]. It is convenient to work on the double cover $T^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$, where ι is the elliptic involution and a curve in P lifts to an ι -invariant family; pillowcase counts are recovered from the cover as in §3.2.

A rational tangle Q_r has for its traceless character variety a straight line of slope r through the corners [4]. Two operations build the curves we need:

- *Conway sum* $T_1 + T_2$. On the pillowcase the sum is a fiber product over the seam coordinate γ : the θ -coordinates add, $[\gamma, \theta_1 + \theta_2]$, and over each seam $\gamma \in \{0, \pi\}$ a circle of reducibles (a *seam fiber circle*) appears when the two summands are both non-abelian there [10, Thm. 4.22]. Perturbing the middle piece resolves each circle by a cut-and-paste that reconnects the four incident branches diagonally [10, Thm. 4.23], producing one new self-intersection per resolved circle.
- *Earring* $\widehat{Q_r}$. Adding the earring tangle replaces the arc of a tangle by its Herald–Kirk figure-eight $\mathcal{R}^\natural$: the doubled arc closing through the corner folds, with one forced self-crossing (the *pinch*) [6, Def. 6.6]. For a rational tangle the pinch supports the trivial bounding cochain [10, Lem. 5.6].

2.2. The two Lagrangians. For the decomposition (2), the -2 band is the same for every k , so one Lagrangian is fixed across the family and only the other varies:

- **red** $= \mathcal{R}^\natural(T_1)$, the earring figure-eight of the line of $Q_{-1/2}$ in the pairing frame—*independent of k* . It is a single immersed circle in P with one self-intersection (the pinch), on which the bounding cochain is 0.
- **blue** $= \mathcal{R}_t(T_2) = \mathcal{R}_t(Q_{1/3} + Q_{1/(2k+1)})$, the perturbed Conway sum. It has two seams, each carrying two fiber circles; the four resolutions make it a single immersed circle in P . For $k = 2$ it has *forty* self-intersection points; for $k = 3$, *eighty-two*.

We built both curves as explicit piecewise-linear ι -invariant families on T^2 from the representation theory—the compass convention and the Conway-sum fiber coordinate, the seam fiber-circle arc $[|\theta_1 - \theta_2|, \pi - |\pi - \theta_1 - \theta_2|]$, the diagonal resolution, and the earring closure through the corner folds—each step checked against the corresponding figure and formula of [10, 6]; the details and validation batteries are in the source. Figure 1 shows the $k = 2$ pair, the pillowcase analogue of Smith’s Figure 31.

2.3. The validation gates. The purpose of the reconstruction is that it can be checked before it is used.

$P(-2, 3, 5)$. Red and blue meet in nine points, which on P are one interior intersection near $(1.61, 0.86)$ and two clusters of four near the corner column $\gamma = \pi$, at $\theta \approx 1.52$ and $\theta \approx 4.72$; this is Smith’s count. A winding-number search (§3) finds exactly two immersed bigons with all four vertices distinct, so the differential has rank 2 and

$$\text{rank } HF((\mathcal{R}^\natural, 0), (\mathcal{R}_t, 0)) = 9 - 2 \cdot 2 = 5.$$

This agrees with [10, Thm. 5.9] in every entry (nine points, two bigons with distinct vertices, rank 5), and since $\text{rank } I^\natural(P(-2, 3, 5)) = 7$ [10, computed from Khovanov homology and the Alexander polynomial] it re-proves Smith’s conclusion $b \neq 0$.

$P(-2, 3, 7)$. Here Smith gives no numbers to match; the gate is internal consistency. The generic complex has thirteen intersection points and, by the same search, three bigons (differential rank 3), naive rank 7. (The generator count is sensitive to a non-generic choice of the earring

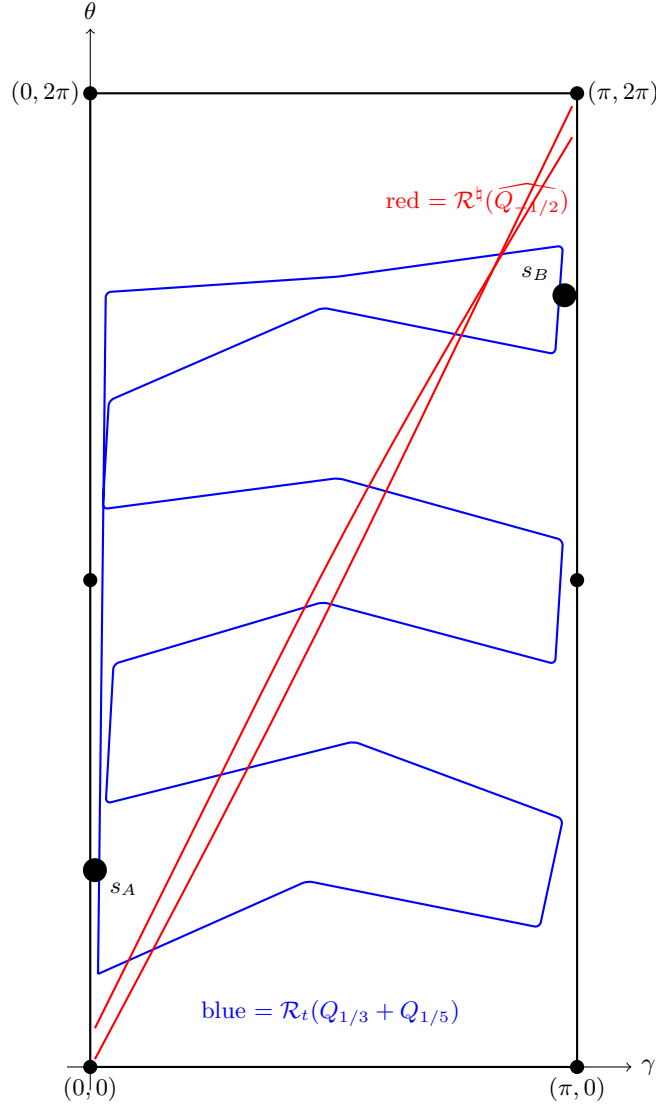


FIGURE 1. Schematic of the two Lagrangians for $P(-2, 3, 5)$ on the fundamental domain $[0, \pi] \times [0, 2\pi]$ of the pillowcase (Smith’s Figure 31). Blue is the perturbed Conway sum $\mathcal{R}_t(Q_{1/3} + Q_{1/5})$, an immersed circle with forty self-intersections clustered along the two seams $\gamma \in \{0, \pi\}$; red is the earring figure-eight $\mathcal{R}^h(\widehat{Q_{-1/2}})$. The marked points s_A, s_B are the support of the computed bounding cochain $b = s_A + s_B$; they lie on opposite seams and are exchanged by $(\gamma, \theta) \mapsto (\pi - \gamma, -\theta)$. The drawing is schematic; the counts below are exact.

perturbation— one degenerate choice collapses it to nine—so we fix a generic perturbation and confirm the count is stable across perturbations; see §5.1.) The target is $\text{rank } I^h(P(-2, 3, 7)) = 9$, the sum of absolute values of the coefficients of the Alexander polynomial of $P(-2, 3, 7)$ (a genus-5 knot), which is the $l(K) = u(K)$ value in Smith’s sense. Everything past these gates is new.

3. THE COMBINATORIAL FLOER PACKAGE

3.1. Bigons by winding numbers. For two immersed curves in a surface, the Floer differential counts immersed bigons, which de Silva–Robbin–Salamon detect combinatorially by a winding function [3]. We use the pillowcase version. Fix generators x, y (intersection points of red and blue). A candidate bigon from x to y is a loop $\Gamma = \alpha \cdot \beta$ formed by an arc α of red from x to y and an arc β of blue from y to x . Lift Γ to T^2 and let $w: T^2 \setminus \Gamma \rightarrow \mathbb{Z}$ be its winding (intersection) function. Then Γ bounds an immersed bigon iff

- (a) $[\Gamma] = 0 \in H_1(T^2)$;
- (b) $w \equiv 0$ at the four corners of P (the disk avoids the orbifold points, which also fixes the additive constant);
- (c) $w \geq 0$ on every complementary region; and
- (d) the two corners of Γ at x and y are convex, i.e. each is a single embedded sector of angle $< \pi$ subtending winding jump $+1$.

The differential is $\partial x = \sum_y (\#\{\text{bigons } x \rightarrow y\} \bmod 2) y$, and $\text{rank } HF = \#\{\text{generators}\} - 2 \text{rank } \partial$ over $\mathbb{F}_2$. Conditions (a)–(d) are exactly what our program tests, by adaptive sampling of w on both sides of every edge and a sector test at each corner; the delicate points (long edges under mod- 2π arithmetic, shallow-angle corners) are handled as recorded in the source.

3.2. The lift bookkeeping. Because we count on the double cover, we must be sure the cover count equals the pillowcase count. A pillowcase polygon lifts to exactly two ι -related polygons on T^2 ; fixing one lift of red (say the component C_+ of the earring) selects exactly one lift of each pillowcase polygon whose red arc lies on C_+ . Hence

$$\#\{\text{pillowcase polygons}\} = \#\{T^2 \text{ polygons with red arc on } C_+\},$$

and no division is needed. This is the same device by which the bigon count of §2.3 matches Smith’s; it extends verbatim to the higher polygons below. A pillowcase self-intersection of blue has two T^2 preimages, and a pillowcase polygon through it may close through either; we therefore take the union over preimages when tabulating a self-crossing’s polygon contributions. *One pitfall we flag:* some generators fold under $T^2 \rightarrow P$ (they sit near $\theta \approx 1.5$ on the cover but map near $\theta \approx 4.7$ on P), so proximity on P is *not* proximity on the cover; all locality tests must be phrased on the cover (by blue arc-length between anchor points), not in pillowcase coordinates. Ignoring this drops genuine polygons, and was, in an intermediate version of this work, the source of a false negative that we correct here.

3.3. The deformed differential. Let $\mathcal{R}$ be an immersed Lagrangian with self-intersections and $b = \sum_i c_i s_i$ ($c_i \in \mathbb{F}_2$) a bounding cochain on it. Following [2, §2.9] and [10, §5], the differential of the pair $((\mathcal{R}^\natural, 0), (\mathcal{R}, b))$ counts immersed polygons two of whose vertices are generators (intersection points of red and blue) and each of whose remaining vertices is a self-intersection s_i in the support of b :

$$\partial_b(x) = \sum_y \left(\#\mathcal{B}(x, y) + \sum_i c_i \#\mathcal{T}(x, s_i, y) + \sum_{i < j} c_i c_j \#\mathcal{Q}(x, s_i, s_j, y) + \cdots \right) y \pmod{2}, \quad (3)$$

where $\mathcal{B}$ are bigons, $\mathcal{T}$ triangles (one b -vertex), $\mathcal{Q}$ quadrilaterals (two b -vertices), and so on; these are the terms of the A_∞ maps $\mu^1, \mu^2, \mu^3, \dots$ [10, §5.2]. Each polygon is a loop whose boundary alternates: an arc of red between the two generators, and arcs of blue between consecutive vertices, switching blue branch at each self-crossing. The very same winding conditions (a)–(d)—now with a convex-corner test at *every* vertex, at x and y (red $\cap$ blue corners) and at each s_i (blue $\cap$ blue corners)—decide whether the loop bounds an immersed disk. We validated the polygon counter

on the unlink prototype of [10], where a single triangle through the earring pinch is the mechanism of that example.

The Maurer–Cartan equation (1) is the analogous count with *all* vertices at self-intersections: for each self-crossing s ,

$$\left[\sum_k \mu^k(b, \dots, b) \right]_s = \# \mathcal{M}(s) + \sum_i c_i \# \mathcal{B}^{\mathcal{R}}(s_i, s) + \sum_{i,j} c_i c_j \# \mathcal{T}^{\mathcal{R}}(s_i, s_j, s) + \dots = 0, \quad (4)$$

where $\mathcal{M}$ are monogons (one-cornered teardrops, the curvature μ^0), $\mathcal{B}^{\mathcal{R}}$ blue self-bigons (μ^1), and $\mathcal{T}^{\mathcal{R}}$ blue triangles (μ^2)—all polygons with boundary entirely on blue.

3.4. Unobstructedness. For (1) to be a meaningful equation—rather than a formal series with a nonzero constant term—the curvature μ^0 must vanish. It does, for the standard reason.

Proposition 3.1. *The blue Lagrangians $\mathcal{R}_t(Q_{1/3} + Q_{1/(2k+1)})$ bound no teardrops: $\mu^0 = 0$. They are unobstructed immersed Lagrangians, and (1) is a genuine Maurer–Cartan equation.*

Proof. A teardrop is a Maslov-index-1 immersed disk with a single boundary corner, so the curvature $\mu^0 \in CF(\mathcal{R}, \mathcal{R})$ lies in degree 2. For an immersed curve in a surface, a transverse self-intersection contributes two generators, of degrees d and $1 - d$ (Akaho–Joyce [1]); hence $CF(\mathcal{R}, \mathcal{R})$ is concentrated in degrees 0 and 1, and $CF^2(\mathcal{R}, \mathcal{R}) = 0$, so $\mu^0 = 0$. (The argument requires the disks to lie in the smooth stratum P^* , where a $\mathbb{Z}$ -grading is available; we verified computationally that every self-intersection of blue lies at distance ≥ 0.6 from every corner, so all teardrop candidates do. The monogon count of §4.1 confirms $\mu^0 = 0$ directly, and the counter was checked to detect a genuine teardrop on a test curve.) $\square$

This is the standard mechanism by which graded curves in surfaces are unobstructed; we include it because it is what makes the deformed theory well posed, and because it is easy to lose track of in the immersed setting.

4. THE COMPUTATIONS

We prove Theorem 1.1. The gates are §2.3; here we compute the Maurer–Cartan data (§4.1), then the two bounding cochains (§4.2, §4.3).

4.1. Maurer–Cartan data. By Proposition 3.1, $\mu^0 = 0$ for both blue curves. The higher self-Floer products differ between the two members, and we record the honest picture.

$P(-2, 3, 5)$. All three counts of (4) vanish: no monogons ($\mu^0 = 0$), no self-bigons ($\mu^1 = 0$; every ordered pair of self-crossings within a blue-arc window was tested), and no self-triangles ($\mu^2 = 0$; all $\binom{40}{3}$ triples were tested, window-pruned). Hence for any $b = \sum c_i s_i$ with $|\text{supp } b| \leq 2$, (4) reduces to 0: the μ^0, μ^1, μ^2 terms vanish, and the first potentially surviving term $\mu^3(b, b, b)$ counts blue quadrilaterals with three vertices in $\text{supp } b$, impossible for a two-element support (an immersed quadrilateral has four distinct vertices). So *Maurer–Cartan is vacuous on small support*, and the bounding cochain is pinned entirely by the rank condition. It also means $b = 0$ is a valid bounding cochain, consistent with Smith computing rank 5 there.

$P(-2, 3, 7)$. Again $\mu^0 = 0$, but now $\mu^1 \neq 0$: the blue curve has *two* self-bigons. So Maurer–Cartan is a real constraint here— b must be μ^1 -closed—not a vacuous one. The bounding cochain we find below is supported on a single crossing that is μ^1 -closed (it is the tail of no self-bigon), so (1) holds; had it not been μ^1 -closed, no single-crossing cochain would be admissible.

4.2. $P(-2, 3, 5)$: the quadrilateral. By §2.3 the naive differential has rank 2, carried by two bigons; write their generator pairs (x_1, y_0) and (x_4, y_6) , with y_0 the interior generator and (x_4, y_6) the pair in the $\theta \approx 4.72$ cluster. We seek b with rank $\partial_b = 1$, so rank $HF = 9 - 2 = 7$.

Lemma 4.1. *Over $\mathbb{F}_2$, a differential extending the rank-2 ∂_0 by polygon terms as in (3) has rank 1 only if those terms cancel exactly one bigon entry and introduce nothing that survives outside the remaining bigon's row.*

Proof. ∂_0 has its two nonzero entries in distinct rows and columns, hence rank 2. Adding an $\mathbb{F}_2$ -matrix can lower the rank to 1 only by cancellation: the sum must have all nonzero rows proportional. Since additions never remove the possibility of raising rank, the outcome is that the added terms cancel one bigon entry and add nothing surviving outside the other bigon's row. $\square$

Some polygon must therefore contribute to (x_4, y_6) or (x_1, y_0) . No triangle does: the single- b -vertex terms of (3) were enumerated over all forty crossings, and none has a triangle on either bigon pair. The cancellation is achieved one order higher.

Proposition 4.2. *There is an immersed quadrilateral with boundary vertices x_4, y_6 and interior vertices two self-crossings s_A, s_B ; it contributes to the entry (x_4, y_6) of (3). Neither s_A nor s_B carries any triangle or monogon, and their only shared quadrilateral is this one. Hence for $b = s_A + s_B$,*

$$\partial_b = \partial_0 + \mathcal{Q}(x_4, s_A, s_B, y_6) \cdot E_{x_4, y_6}$$

has the single entry (x_1, y_0) : the quadrilateral cancels the (x_4, y_6) bigon and nothing else changes. Therefore rank $\partial_b = 1$ and rank $HF = 7$.

Proof (computation). The quadrilateral is found by the polygon counter of §3.3: the loop $\text{red}(x_4 \rightarrow y_6) \cdot \text{blue}(y_6 \rightarrow s_B) \cdot \text{blue}(s_B \rightarrow s_A) \cdot \text{blue}(s_A \rightarrow x_4)$, switching blue branch at s_A and s_B , satisfies (a)–(d) with a convex corner at each of the four vertices, and it is stable as the arc-length window is enlarged over a wide range, so it is a genuine immersed disk and not a pruning artifact. Enumerating the triangle terms shows s_A, s_B carry none, and the full quadrilateral tabulation shows their only shared quadrilateral is the one above. Substituting into (3): the bigon terms give ∂_0 ; the triangle terms vanish on $\{s_A, s_B\}$; the single quadrilateral term adds E_{x_4, y_6} , cancelling that bigon modulo 2. The result has the single entry (x_1, y_0) , rank 1, so rank $HF = 7$. $\square$

Proposition 4.3. *Among all bounding cochains supported on the forty self-crossings, $b = s_A + s_B$ is the unique one of minimal support with rank $HF = 7$: no single crossing achieves it, and $s_A + s_B$ is the only two-crossing support that does.*

Proof (computation). Assembling (3) from the tabulated triangle and quadrilateral terms gives ∂_b as an explicit $\mathbb{F}_2$ -function of the coefficients c_i . A search over supports by increasing size evaluates rank ∂_b : every single crossing gives rank ≥ 2 , and among all $\binom{40}{2}$ pairs exactly one, $\{s_A, s_B\}$, gives rank 1. Every other pair either fails to touch a bigon or drags in a triangle that spoils the cancellation—for instance the geometrically nearest canceling pair (the two crossings flanking the (x_4, y_6) bigon) cancels that bigon but carries triangles $(x_3, y_6), (x_3, y_4), (y_6, y_5)$, leaving rank 3. Minimality and uniqueness follow. By §4.1, b satisfies Maurer–Cartan. $\square$

This proves Theorem 1.1(i); the two support crossings lie near $(0.03, 1.27)$ and $(3.06, 4.98)$, one on each seam, exchanged by $(\gamma, \theta) \mapsto (\pi - \gamma, -\theta)$.

4.3. $P(-2, 3, 7)$: the triangle. Here the naive differential has rank 3 (three bigons), naive rank $HF = 7$; the target is rank $HF = 9$, so we seek b with rank $\partial_b = 2$ —i.e. the deformed differential must cancel exactly one of the three bigons. Unlike the $k = 2$ case, a triangle suffices.

Proposition 4.4. *There is a single self-crossing s of blue, near $(\gamma, \theta) = (0.05, 5.41)$, whose immersed triangle contributes to the chain-level entry of one of the three bigons; s carries no other polygon touching the generators, and s is μ^1 -closed. Hence $b = s$ satisfies Maurer–Cartan and $\partial_b = \partial_0 + \mathcal{T}(\cdot, s, \cdot)$ has rank 2, so $\text{rank } HF((\mathcal{R}^\natural, 0), (\mathcal{R}_t, b)) = 13 - 2 \cdot 2 = 9 = I^\natural$. Moreover $b = s$ is the unique minimal (single-crossing) bounding cochain with this rank.*

Proof (computation). The triangle $\text{red}(x \rightarrow y) \cdot \text{blue}(y \rightarrow s) \cdot \text{blue}(s \rightarrow x)$, with x, y the two generators of one bigon, satisfies (a)–(d) and is window-stable, so it is a genuine immersed disk; its generator pair coincides with a bigon of ∂_0 , so it cancels that bigon. A search over all single crossings, evaluating $\text{rank } \partial_b$ from the tabulated triangle terms and filtering by the Maurer–Cartan check of §4.1, returns exactly one crossing with $\text{rank } \partial_b = 2$, namely s ; s is the tail of no self-bigon, so $\mu^1(b) = 0$ and (1) holds (with $\mu^{\geq 2}(b, \dots) = 0$ automatically for a one-element support). This is the unique minimal solution. $\square$

This proves Theorem 1.1(ii). We note the contrast with $P(-2, 3, 5)$: there, no triangle reached a bigon and the correction was forced up to a quadrilateral; here a triangle reaches a bigon, and one crossing suffices. The +2 correction is the same; the polygon is not.

5. ROBUSTNESS, THE CONJECTURE, AND THE FAMILY

5.1. Stability under perturbation. The immersed curves depend on perturbation parameters (the shear resolving the seam circles, the earring amplitude and phase), and a genuine invariant must not depend on them. We ran the entire pipeline at two independent choices of parameters, for each member. For $P(-2, 3, 5)$: both give nine generators, two bigons, rank 5; both give a unique two-crossing bounding cochain with $\text{rank } HF = 7$, with support crossings at the same pillowcase coordinates (within 0.02). For $P(-2, 3, 7)$: both give thirteen generators, naive rank 7; both give a unique single-crossing cochain with $\text{rank } HF = 9$, with the support crossing at the same coordinate (within 0.02). (One non-generic earring perturbation collapses the $k = 3$ generator count to nine; we exclude it and confirm the count is otherwise stable.) The values are stable.

5.2. What is proved, computed, and conjectured. The reconstruction of the Lagrangians and the deformed-differential formalism are re-implementations of [4, 5, 2, 6, 10], validated against [10] through the rank-5 gate; unobstructedness (Proposition 3.1) is the standard degree fact [1]. The Maurer–Cartan data, the cancelling polygons, and the uniqueness statements are computer computations; the source and its validation batteries are included. That each b is *the* bounding cochain of the corresponding tangle—and hence realizes the Atiyah–Floer counterpart of $I^\natural$ —holds under the standing conjecture of [2] that the immersed-curve model computes the Floer-theoretic invariant, the same hypothesis under which [2, 10] are stated. Smith [10, Thm. 5.9] proves unconditionally that a nonzero bounding cochain exists for $P(-2, 3, 5)$; we compute its value, and that of $P(-2, 3, 7)$, within the model.

5.3. The family beyond $k = 3$. It is natural to ask whether the pattern of Theorem 1.1—naive rank below $I^\natural$, corrected by +2 through a single bigon cancellation—persists along the family (2). We do not know, and the computed data cautions against assuming it. Table 1 collects what we have.

Two cautions. First, the naive rank for $P(-2, 3, 11)$ is 15 (perturbation-stable), which *exceeds* the value one would extrapolate from $q = 5, 7$; if $I^\natural(P(-2, 3, 11)) < 15$, the correction there would have to *lower* the rank—the opposite direction from $k = 2, 3$ —which would be a genuine change of behavior within a single family. We cannot decide this without $I^\natural(P(-2, 3, 11))$, which the small-genus, L -space reasoning available for $q \leq 7$ does not supply. Second, the Maurer–Cartan data already varies ($\mu^1 = 0$ for $k = 2$, $\mu^1 \neq 0$ for $k = 3$), so the clean “rank alone pins b ” of

$q = 2k+1$	knot	genus	naive rank HF	$I^\natural$	b	mechanism
3	$8_{19} = T(3, 4)$	3	5	5	0	— (embedded)
5	$P(-2, 3, 5) = T(3, 5)$	4	5	7	two crossings	μ^3 quadrilateral
7	$P(-2, 3, 7)$	5	7	9	one crossing	μ^2 triangle
11	$P(-2, 3, 11)$	7	15	?	?	?

TABLE 1. The family (2). Bold-faced entries elsewhere in the text are computed here or firmly known; $I^\natural$ for $q \geq 11$ we do not determine. The naive rank 15 for $q = 11$ is perturbation-stable and does not fit the small- q pattern.

$P(-2, 3, 5)$ is not a family-wide phenomenon. What the two computed examples establish is not a family theorem but a proof of concept: once the curves are in hand, computing the bounding cochain reduces to a finite mod-2 search, and the answer can be genuinely small. Whether a uniform statement exists is, in our view, the right open question—and the reduced complex of $P(-2, 3, 2k+1)$, whose $\mathbb{Z}/4$ gradings are worked out in [11], is where its instanton-side shadow should be read.

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